

In-Class Assignment – 01

OOP with Java

Part A: Operators

Q1. [Arithmetic Operators]

Write a program to input three subject marks, calculate the total, average, and percentage, and display the grade as per the following rule:

- Percentage $\geq 75 \rightarrow$ Distinction
- 60–74 $\rightarrow$ Credit
- 50–59 $\rightarrow$ Pass
- Below 50 $\rightarrow$ Fail

Q2. [Unary Operators]

Trace and display the output of the following code snippet (or implement it and verify):

```
int x = 5;
int y = ++x + x++ + --x + x--;
System.out.println("x = " + x + ", y = " + y);
```

Explain how unary operators affected the final result.

Q3. [Logical Operators]

Write a program to check whether a number is divisible by both 3 and 5, either 3 or 5, or neither.

Q4. [If Statement]

Input a number and check if it is positive.

Q5. [If-Else]

Input an integer and check if it is even or odd using if-else.

Q6. [If-Else-If Ladder]

Write a program to categorize a person's BMI value:

- $<18.5 \rightarrow$ Underweight
- 18.5–24.9 $\rightarrow$ Normal
- 25–29.9 $\rightarrow$ Overweight
- $\geq 30 \rightarrow$ Obese

Q7. [Switch Case]

Write a simple calculator program that accepts two numbers and an operator (+, -, *, /, %). Use a switch statement to perform the operation and display the result.

Q8. [For Loop]

Display the multiplication table of any number entered by the user (from 1 to 12).

Q9. [Nested For Loop]

Print the following pattern using nested for loops:

```
1
12
123
1234
12345
```

Q10. [While Loop]

Write a program to sum all even numbers between 1 and 100 using a while loop.

Q11. [Do-While Loop]

Keep asking the user to enter a number until they enter a negative number. Finally, display how many positive numbers were entered.

Q12. [Break Statement]

Write a loop from 1 to 100. Stop the loop when the number reaches 50 and print all numbers before it.

Q13. [Continue Statement]

Display numbers from 1 to 20, but skip all multiples of 3 using the continue statement.

Q14. [Combined Operators & If]

Write a program to input three integers and display the largest number using only if-else statements and logical operators.

Q15. [Loop + Conditional + Break]

Simulate a basic ATM withdrawal system:

- Initial balance = 10,000
- User can withdraw multiple times
- Stop when balance < 500 or user chooses to exit
- Use a do-while loop with break and continue where appropriate.